FIFA World Cup 2026 Dashboard

Tournament Stage Sorting and Relationship Configuration in Power BI



1. Introduction

As part of the FIFA World Cup 2026 Dashboard project, a dedicated dimension table was created in Power BI to organize tournament stages in their correct sequence. A relationship was established between the dimension table and the match data table to support accurate data organization and visualization.

This implementation focuses on the 2026 Match Timeline visual, ensuring that tournament stages appear in the intended order, from the Group Stage through the Final, rather than being sorted alphabetically.

2. Objectives

The main objectives of this implementation are:

- To create a dedicated dimension table containing tournament stages and their corresponding sort-order values.
- To define the correct sequence of tournament stages.
- To establish a relationship between the stage dimension table and the match data table.
- To configure the stage column to display tournament stages in the correct order.
- To improve the structure, consistency, and maintainability of the Power BI data model.

3. Creating the Stage Dimension Table

3.1 Purpose of the Dimension Table

In Power BI, categorical values are often sorted alphabetically by default. For a tournament timeline, alphabetical sorting does not represent the actual sequence of competition stages.

To address this issue, a separate dimension table named Dimension Stage was created. This table contains two columns:

- **Stage:** Stores the names of the tournament stages.
- **SortOrder:** Stores a numeric value defining the display order of each stage.

The SortOrder column provides a reference that Power BI can use to display tournament stages in the intended sequence.

3.2 Procedure for Creating the Table

1. Open the FIFA World Cup 2026 Dashboard project in Power BI Desktop.
2. Navigate to the Modeling tab on the ribbon.
3. Select New table.
4. Enter the following DAX expression in the formula bar.

DAX Formula:

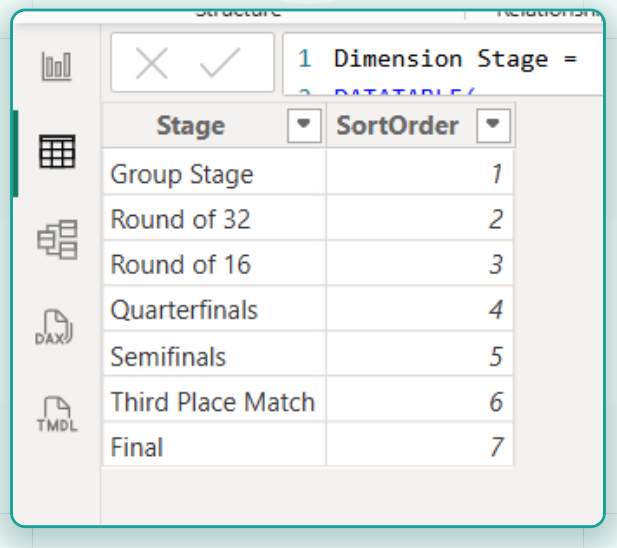
```
Dimension Stage =
DATATABLE(
  "Stage", STRING,
  "SortOrder", INTEGER,
  {
    {"Group Stage", 1},
    {"Round of 32", 2},
    {"Round of 16", 3},
    {"Quarterfinals", 4},
    {"Semifinals", 5},
    {"Third Place Match", 6},
    {"Final", 7}
  }
)
```

5. Press Enter to create the calculated table.
6. Verify that the **Dimension Stage** table appears in the Data or Table view.

> **Note:** The table stores predefined tournament stages and their corresponding sort-order values.

3.3 Screenshot: Stage Dimension Table

Capture a screenshot in Power BI Desktop showing the **Dimension Stage** table with both columns, **Stage** and **SortOrder**, clearly visible.



Stage	SortOrder
Group Stage	1
Round of 32	2
Round of 16	3
Quarterfinals	4
Semifinals	5
Third Place Match	6
Final	7

4. Establishing the Relationship Between the Tables

4.1 Purpose of the Relationship

The Dimension Stage table must be connected to the match data table so that tournament stages can be used to organize and filter match records.

The relationship connects the tournament-stage column in the FIFA World Cup 2026 Match table to the corresponding Stage column in the Dimension Stage table.

Since multiple matches can belong to the same tournament stage, the match table contains repeated stage values, while the dimension table contains each stage only once. Therefore, a many-to-one (*:1) relationship is appropriate.

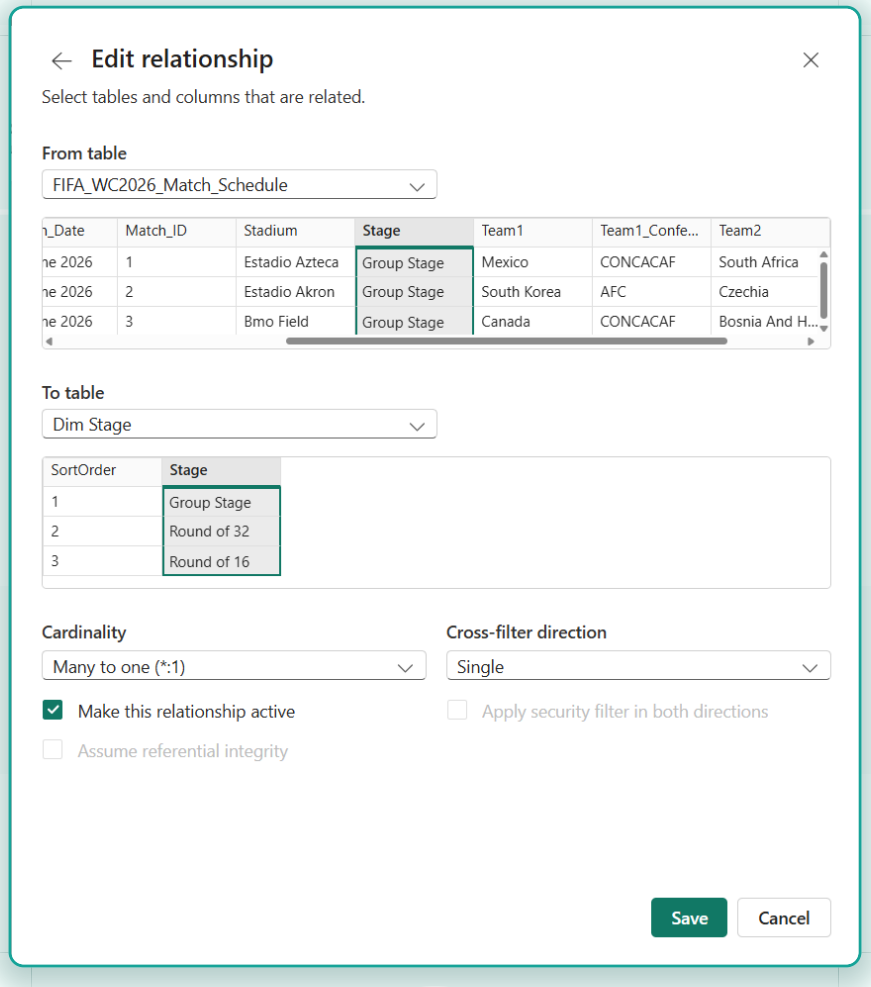
4.2 Procedure for Creating the Relationship

1. Navigate to Model view in Power BI Desktop.
2. Locate the FIFA World Cup 2026 Match table and the Dimension Stage table.
3. Identify the column in the match table that contains the tournament-stage names.
4. Drag the stage column from the FIFA World Cup 2026 Match table onto the Stage column in the Dimension Stage table.

   Alternatively, navigate to Modeling → Manage relationships → New to create the relationship through the relationship configuration dialog. The exact ribbon layout may vary depending on the Power BI Desktop version.

5. Configure the relationship using the following settings:
   - Primary table: FIFA World Cup 2026 Match
   - Primary column: The column containing tournament-stage values
   - Related table: Dimension Stage
   - Related column: Stage
   - Cardinality: Many to one (*:1)
   - Cross-filter direction: Single
   - Make this relationship active: Selected
6. Confirm the configuration and create the relationship.
7. Verify that the relationship appears in Model view and is active.

> **Important:** The stage values in both tables must match exactly, including spelling and spacing. The Stage column in Dimension Stage must contain unique values to serve as the one-side of the relationship.



4.3 Expected Relationship Structure

FIFA World Cup 2026 Match (*) ——— (1) Dim Stage

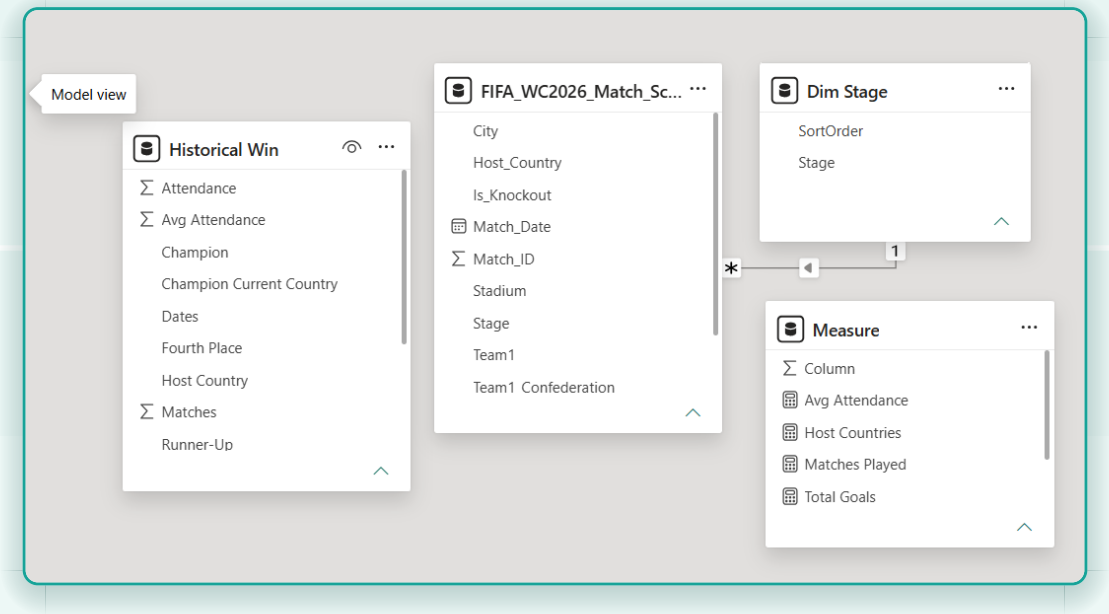
- Many side (*): **FIFA World Cup 2026 Match**
- One side (1): **Dimension Stage**

This relationship allows the dimension table to filter the match data by tournament stage and provides a structured reference for the dashboard visuals.

4.4 Screenshot: Relationship Between the Tables

Capture a screenshot showing both tables and the relationship line connecting the stage column in the match table to the **Stage** column in **Dimension Stage**.

Ensure that the * symbol appears on the match table side and the 1 symbol appears on the dimension table side.



5. Configuring the Tournament Stage Sort Order

5.1 Purpose of the Sort Configuration

Creating the dimension table and establishing the relationship are not sufficient on their own to guarantee the correct display order. Power BI must also be instructed to sort the Stage column according to the numeric values in **SortOrder**.

This configuration ensures that the tournament stages appear in the intended sequence rather than alphabetically.

5.2 Procedure for Configuring the Sort Order

1. Navigate to Data view in Power BI Desktop.
2. Select the **Dimension Stage** table.
3. Select the **Stage** column.
4. Open the Column tools tab on the ribbon.
5. Select Sort by column.
6. Choose **SortOrder** from the available columns.

After this configuration, Power BI will use the values in **SortOrder** whenever it sorts the **Stage** column.

> **Note:** Each stage must correspond to exactly one sort-order value. The **Stage** and **SortOrder** columns must maintain a consistent mapping for the configuration to work correctly.

6. Applying the Sort Order to the Match Timeline Visual

6.1 Procedure for Updating the Visual

After configuring the sort order, the Match Timeline visual must use the stage field from the dimension table.

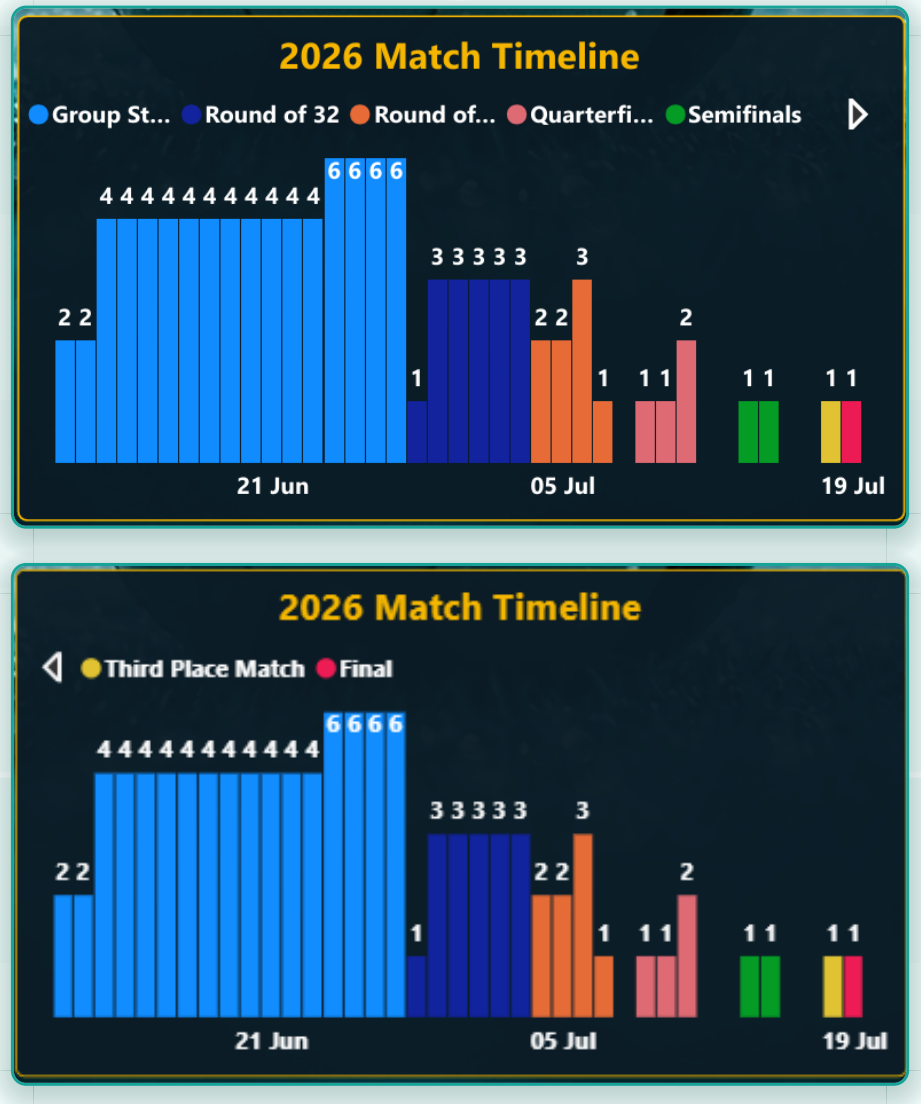
1. Navigate to Report view in Power BI Desktop.
2. Select the Match Timeline visual.
3. In the visual's field configuration, locate the field used to represent tournament stages.
4. Replace the stage field from the match table with **Dimension Stage[Stage]** wherever the visual uses a tournament-stage category or axis.
5. If the visual provides sorting options, select the stage field and set the sort direction to ascending.
6. Review the visual and verify that the stages appear in the intended sequence.

Using **Dimension Stage[Stage]** allows the visual to benefit from the sorting configuration applied to the dimension table.

> **Note:** If the Match Timeline uses a custom visual or a date-based axis, its behavior may depend on the visual's specific configuration. Always verify the final result in Report view.

6.2 Screenshot: Final Match Timeline Visual

Capture the updated Match Timeline visual from the FIFA World Cup 2026 Dashboard, showing the tournament stages in the correct order.



7. Verification and Validation

After completing the table creation, relationship configuration, and sorting setup, the following checks should be performed to ensure that the implementation works correctly.

- Verify that all seven tournament stages are present in the **Dimension Stage** table.
- Confirm that the **SortOrder** column contains the correct integer values from 1 to 7.
- Ensure that the **Stage** column is configured to sort by **SortOrder**.
- Confirm that the relationship between the two tables is active.

- Verify that the match table is on the many side and **Dimension Stage** is on the one side.
- Check that the stage names match exactly in both tables.
- Confirm that the Match Timeline visual uses the stage field from the dimension table.
- Verify that the stages appear in the intended sequence.
- Test the visual's filtering behavior to ensure that the relationship works as expected.

These checks help ensure that the data model and the Match Timeline visual function consistently.

8. Results and Benefits

The creation of the Dimension Stage table and its relationship with the match data table provides several benefits to the FIFA World Cup 2026 Dashboard:

- **Correct stage ordering:** Tournament stages are displayed according to the predefined sequence rather than alphabetical order.
- **Improved data organization:** Tournament-stage definitions are maintained separately from individual match records.
- **Consistent filtering:** The dimension table can be used to filter related match data by stage.
- **Improved maintainability:** The sorting logic is centralized in a dedicated dimension table.
- **Better visualization:** The Match Timeline presents the tournament progression in a more logical and understandable format.

9. Conclusion

This implementation demonstrates how dimension tables, relationships, and custom sorting can be used to improve data organization and visualization in Power BI.

By creating the Dimension Stage table, establishing an active many-to-one relationship, and configuring the Stage column to sort by SortOrder, the FIFA World Cup 2026 Dashboard gains a structured approach to displaying tournament stages in the intended sequence.

This improves the clarity of the Match Timeline visual and establishes a more organized foundation for analyzing and presenting tournament match data.

10. Tools and Concepts Used

Tool: Microsoft Power BI Desktop

Key concepts:

- DAX calculated tables
- DATATABLE() function
- Dimension table creation
- Data modeling and table relationships
- Many-to-one (*:1) cardinality
- Cross-filter direction
- Sort by column
- Report visualization and validation